

PROJECT REPORT

Mandi-to-Market Supply Chain Optimizer

From data rescue to evidence-based supply-chain analysis

SQL arrival volume	Below supplied MSP	Usable transit trips
6.078 million qtl	40.16% of eligible records	9,715 / 10,000

Executive summary

Eight SQL analyses examine crop supply, price exposure and transport duration across synthetic agricultural records. Crop volumes are broadly balanced: Mustard leads at 17.19%, while the five largest mandis account for only 9.62% of valid arrivals. August has the highest observed full-month volume.

Below-benchmark prices occur in 3,667 of 9,131 eligible observations. Wheat has the highest crop-level rate (41.78%), and Nashik Mandi / Rice leads the screened mandi-crop ranking (68.18%, 22 eligible records). Warehouse mean transit times are tightly grouped, so route-level review is more informative than broad destination rankings.

These findings support monitoring and investigation, not claims of financial loss, missed delivery commitments or causal weather effects. Dates, source coverage and synthetic-data limitations remain explicit.

Team and responsibilities

Team member	Contribution
Rohan Khanna	Data cleaning and validation
Bhavini raghuram	SQL modelling and analysis
Niya Avanthika Bijesh	Power BI dashboard
Anamika K C	Report preparation

Scope: eight supplied SQL result sets and six report charts; 2026 arrivals/prices through 9 September. SQL code and full tables remain outside this PDF.

GitHub: github.com/Rohan-sleeps/Agriculture-supply-chain-optimizer

Data, cleaning and methodology

The project addresses fragmented mandi records with multilingual crop names, mixed units, inconsistent IDs and incomplete timestamps. Python cleaning feeds relational SQL analysis and a Power BI reporting layer.

Dataset	Raw rows	Retained	Principal treatment
Master	60	57	3 duplicates removed; 4 states recovered
Arrivals	25,750	25,000	750 duplicates removed; units converted to quintals
Prices	12,000	12,000	Currency parsing; crop and location standardization
Transport	10,400	10,000	400 duplicates removed; distance and duration recovery
Weather	15,000	15,000	Celsius, mm and IST conversion; conflict flags

Analytical method

The supplied schema and queries use MySQL syntax, with CTEs, window functions and relational keys. Fact tables remain separate; mandi joins use the unique master. Medians use ordered middle observations. SQL result CSVs, rather than screenshot estimates, provide the numerical evidence.

Arrivals use valid nonnegative quintal quantities. Price comparisons require positive modal price and supplied MSP in the same record. The price-risk ranking requires at least 20 eligible observations; the route ranking requires at least 15 usable trips. Thresholds are descriptive screening choices, not statistical guarantees.

Cleaning choices that affect interpretation

Missing values stay missing; invalid quantities are not converted to zero. Arrivals retain 1,233 negative-source quantity records outside volume totals. Prices retain 2,377 missing MSP values and 8,366 source/master district conflicts. The master supplies reporting districts; 2,004 price records remain without a reporting district.

Transit duration prioritizes timestamp calculations (7,359 trips), then labelled reported fallbacks (2,356). Weather is cleaned but excluded from the eight business analyses because sensor-location mapping and rainfall intervals are unresolved.

Reconciliation and chart provenance

Q1 and Q2 both sum to 6,077,592 qtl; the earlier Python control was 6,077,460.14 qtl (difference 131.86 qtl, about 0.0022%). Its cause is unverified.

A broadly balanced crop mix

Question: Which crops contribute the most valid arrival volume?

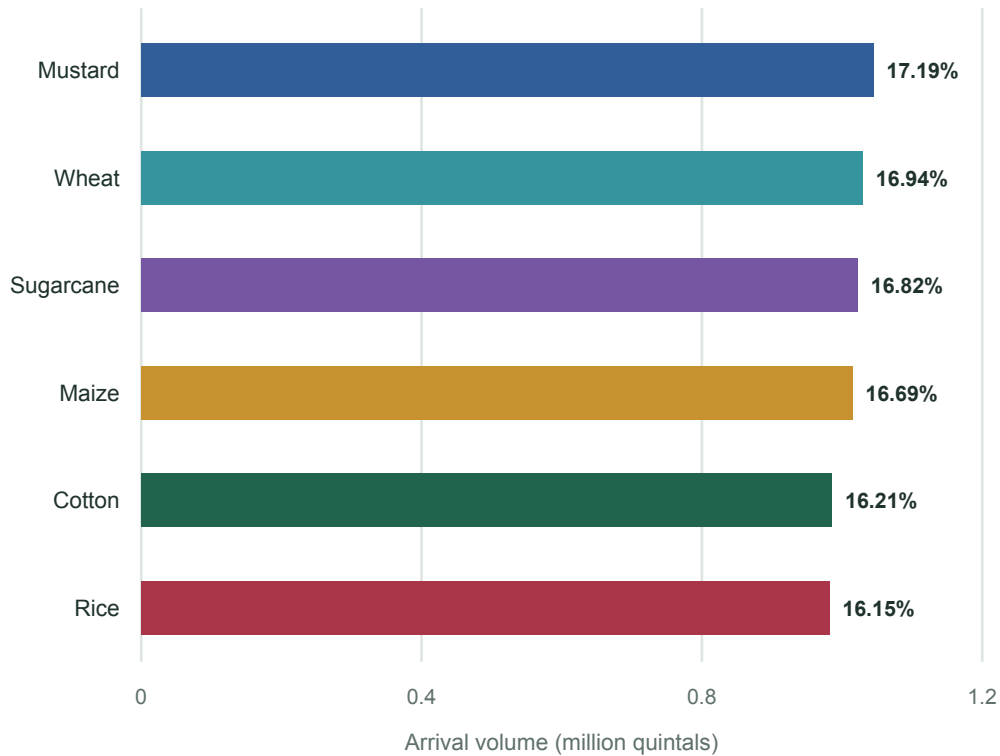


Figure 1. Valid arrival volume by crop. Source: Query 1 result; 23,767 eligible records, January to 9 September 2026. Bar labels show volume share.

Finding: Mustard leads with 1,044,470 qtl (17.19%), while Rice is lowest at 981,673 qtl (16.15%).

Meaning: The narrow 1.04-percentage-point share range suggests broad multi-crop handling requirements rather than one dominant commodity.

Limitation: Volumes exclude invalid quantities, cover synthetic observations and use broad crop groups that do not establish variety equivalence.

Report interpretation

The top three crops - Mustard, Wheat and Sugarcane - together contribute 50.95% of arrivals. Wheat has more valid quantity observations than Mustard (4,074 versus 4,024), illustrating why record counts and volume rankings should not be treated as interchangeable.

Method: sum cleaned arrival_quantity_qtl by crop; divide by the total across eligible crops. Source file: Crop-wise arrival distribution.csv. Full results and SQL belong in GitHub.

August leads the full-month totals

Question: When do arrivals rise or fall across crops?

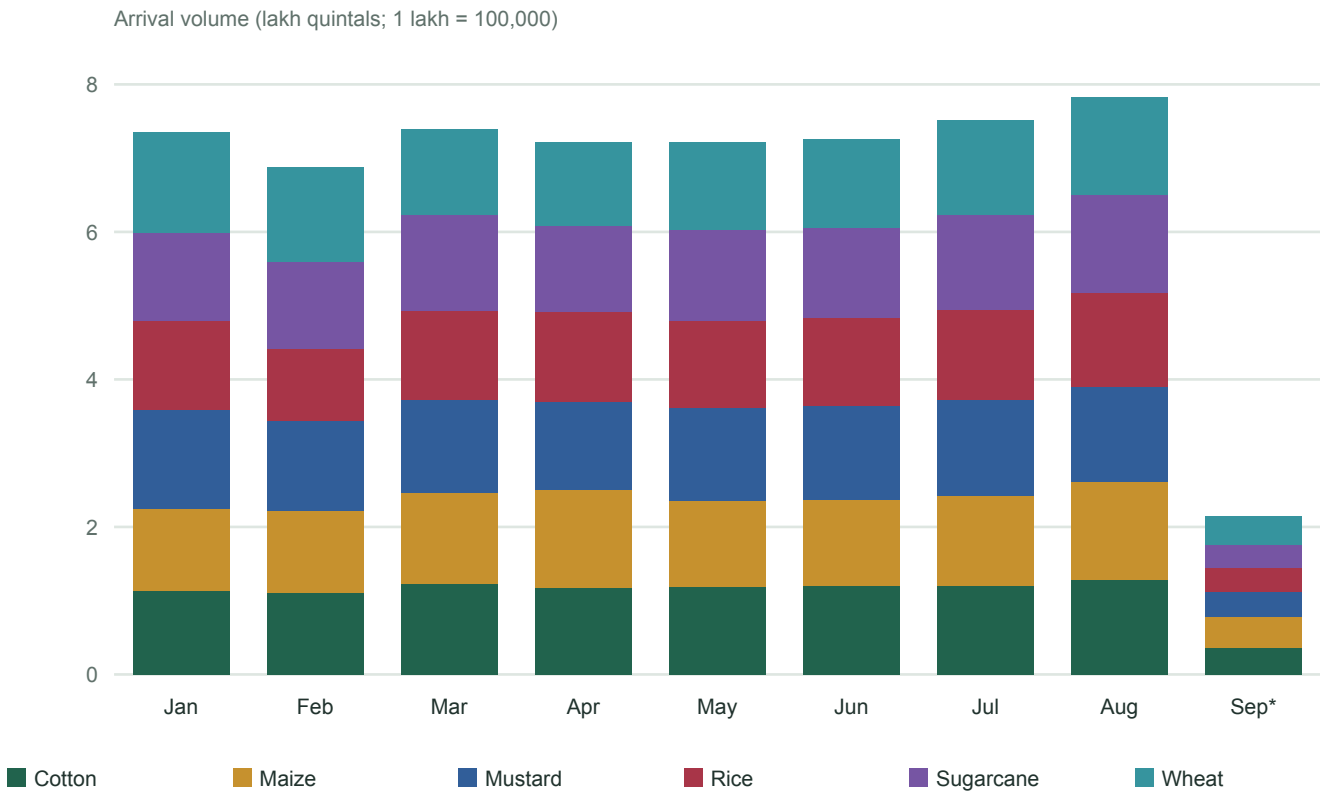


Figure 2. Stacked monthly valid arrivals by crop. Source: Query 2. September* covers only 1-9 September; its shorter coverage prevents a full-month comparison.

Finding: August totals 783,037 qtl, the highest full month, compared with 687,214 qtl in February; March Rice arrivals rise 24.74% from February.

Meaning: Observed monthly variation can guide scenario planning for handling capacity and crop-specific procurement reviews.

Limitation: One synthetic January-September series does not establish recurring seasonality, and September is partial while date assumptions remain flagged.

Changes worth investigating

Total arrivals increase about 4.19% from July to August. Wheat follows a different path: its largest monthly volume is January (135,921 qtl), whereas Sugarcane peaks in August (134,192 qtl). Planning should therefore examine crop-level patterns alongside the aggregate.

The SQL withholds month-on-month changes for partial or nonconsecutive periods. Date coverage does not prove complete observation capture within a month.

Volume is distributed across mandis

Question: How concentrated is supply among the largest mandis?

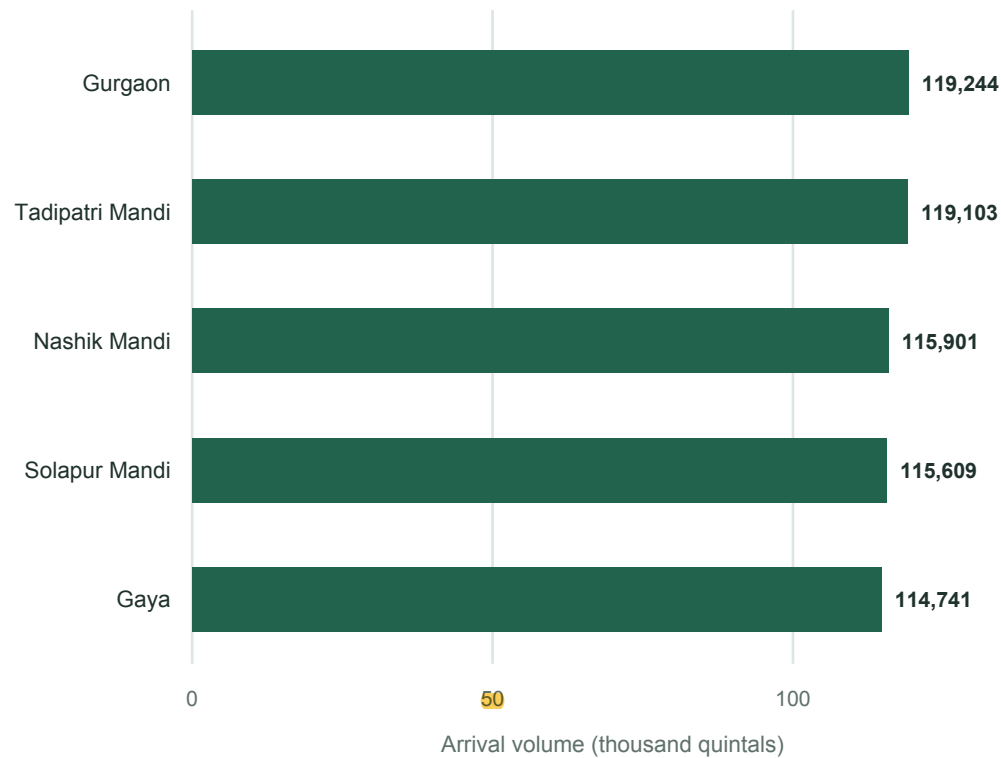


Figure 3. Five largest mandis by valid volume. Source: Query 3; full supplied arrival period. Ranking uses mandi IDs, not names alone.

Finding: Gurgaon Grain Market (MANDI041) leads at 119,244 qtl, narrowly ahead of Tadipatri Mandi (119,103 qtl); the top five contribute 9.62%.

Meaning: The top-five concentration is low, so a network-wide operating approach may matter more than focusing exclusively on the largest sites.

Limitation: Synthetic names are not verified geography, and the 141-qtl gap between the top two is too small to imply a meaningful operational advantage.

Master-table context

The leading two mandis are assigned to Haryana in the master, while Nashik, Solapur and Gaya are assigned to Punjab. These are the supplied synthetic locations; actual place names must not override the reference table.

All arrival mandi IDs matched the master; totals are aggregated before the one-to-one master lookup. Combined top-five volume: 584,598 qtl.

Price risk occurs across all six crops

Question: Which crops most frequently fall below their supplied benchmark?

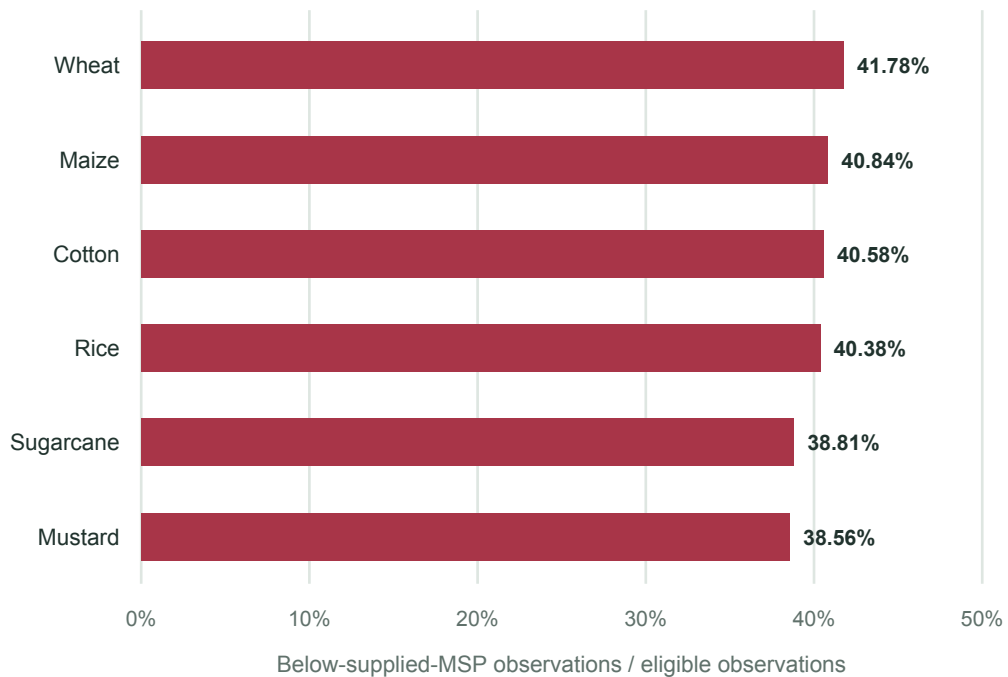


Figure 4. Eligible-observation rates, not transactions or unique-mandi rates. Source: Query 4. Comparison requires both positive modal price and supplied MSP.

Finding: Wheat has the highest rate at 41.78% (630/1,508), while Mustard is lowest at 38.56% (588/1,525); the overall rate is 40.16%.

Meaning: Benchmark exposure is widespread across crops, so monitoring should combine crop-wide coverage with targeted mandi-level review.

Limitation: Missing comparisons exclude 2,869 records, and the supplied price quantity basis and official benchmark applicability are unverified.

Coverage and severity

Comparison coverage ranges from 74.98% for Sugarcane to 77.17% for Rice. Maize has the largest absolute below-benchmark count (649), even though Wheat has the highest rate. All crop-average price gaps are positive, showing that an above-benchmark mean can coexist with many below-benchmark observations.

The overall rate is 3,667 / 9,131, not an average of crop percentages. Currency is INR; the common quantity basis is implied by the challenge.

Targeted mandi-crop review priorities

Question: Which qualifying mandi-crop combinations have the highest below-benchmark frequency?

Selected top five of the ten returned combinations. Eligibility threshold: at least 20 positive modal-price/MSP pairs. Full result table remains in GitHub.

Mandi / crop	Below / eligible	Rate	Mean shortfall*
Nashik Mandi MANDI007 / Rice	15 / 22	68.18%	125.84
Khandwa Grain Market MANDI004 / Sugarcane	14 / 22	63.64%	163.10
Gurgaon Grain Market MANDI041 / Sugarcane	18 / 30	60.00%	151.35
Bilaspur Mandi MANDI056 / Cotton	15 / 25	60.00%	256.14
Nangloi Jat Mandi MANDI031 / Maize	12 / 20	60.00%	116.53

Table 1. Query 5 shortlist. *INR on the supplied basis; shortfall = MSP minus modal price, averaged only over below-benchmark observations.

Finding: Nashik Mandi / Rice leads with 15 of 22 eligible observations below supplied MSP (68.18%), followed by Khandwa Grain Market / Sugarcane at 63.64%.

Meaning: These combinations provide a manageable review list where frequency, sample size and conditional shortfall can be assessed together.

Limitation: This is a selected top-ten ranking with small groups, excludes unmatched mandi IDs and does not establish significance, causality or financial loss.

Frequency and shortfall answer different questions

Among the displayed combinations, Bilaspur Mandi / Cotton has the largest conditional mean shortfall (256.14), but its rate is 60.00%. Gurgaon Grain Market / Sugarcane also has a 60.00% rate, based on 30 eligible observations rather than 25. A review should retain these denominators instead of ranking on percentage alone.

These are investigative priorities in synthetic data. Crop price levels differ, and a larger absolute shortfall across commodities should not automatically be interpreted as greater economic damage.

An above-MSP mean can conceal exposure

Question: How does average Wheat pricing compare with its supplied benchmark?

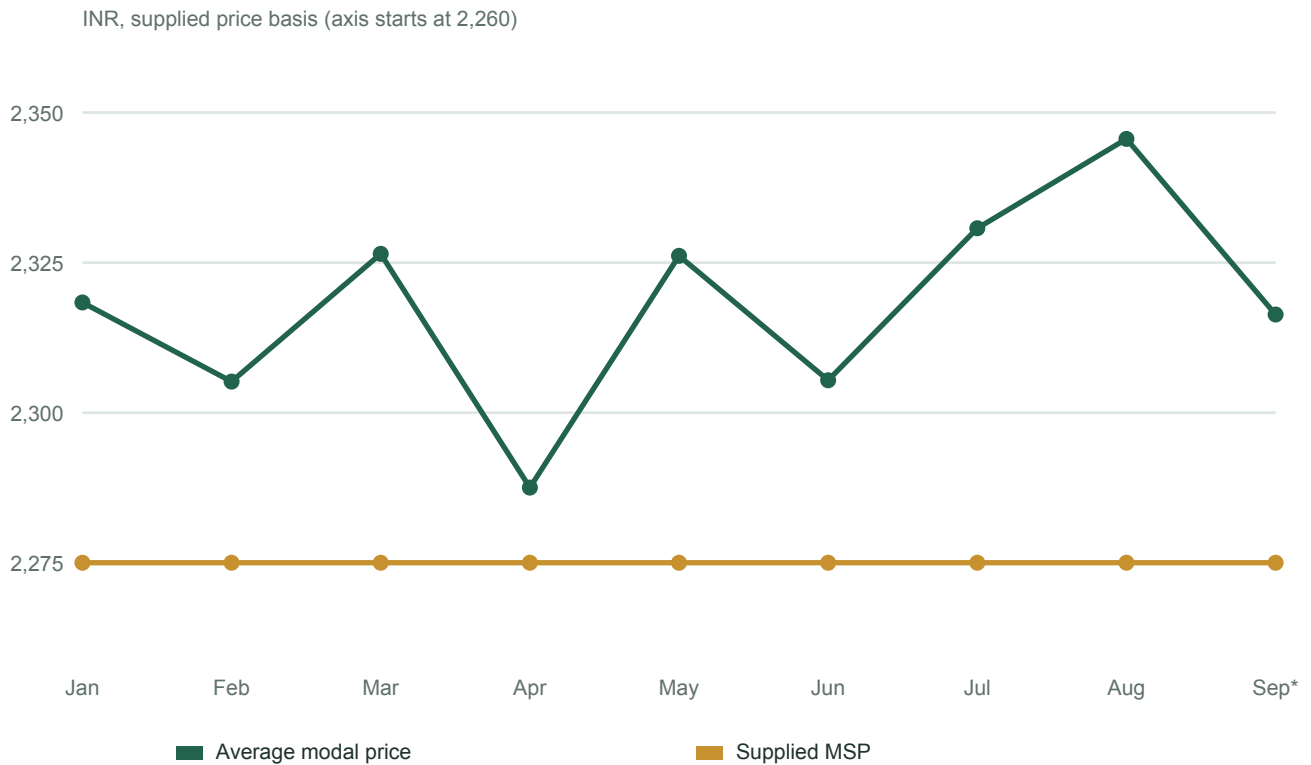


Figure 5. Matched eligible-observation averages for Wheat. Source: Query 6; 1,508 records. September* is partial. This is not the excluded date-sensitivity chart.

Finding: Wheat average modal price stays above the supplied MSP of 2,275 in every month, ranging from 2,287.52 in April to 2,345.61 in August.

Meaning: Monthly averages should be paired with below-benchmark rates because April still has 48.86% of eligible Wheat observations below MSP.

Limitation: The series is unweighted, uses only observations with both values, includes flagged date interpretations and assumes a common supplied price basis.

The lowest average does not mean every price is low

April has a mean premium of only 12.52 and the highest monthly Wheat below-MSP rate in this result. August has a mean premium of 70.61 and a lower rate of 36.90%. These two views distinguish the centre of the distribution from the share of observations below the benchmark.

Both plotted averages use the same monthly eligible population. No arrival-weighted prices, revenue estimates or official-rate claims are made.

Destination averages hide route variation

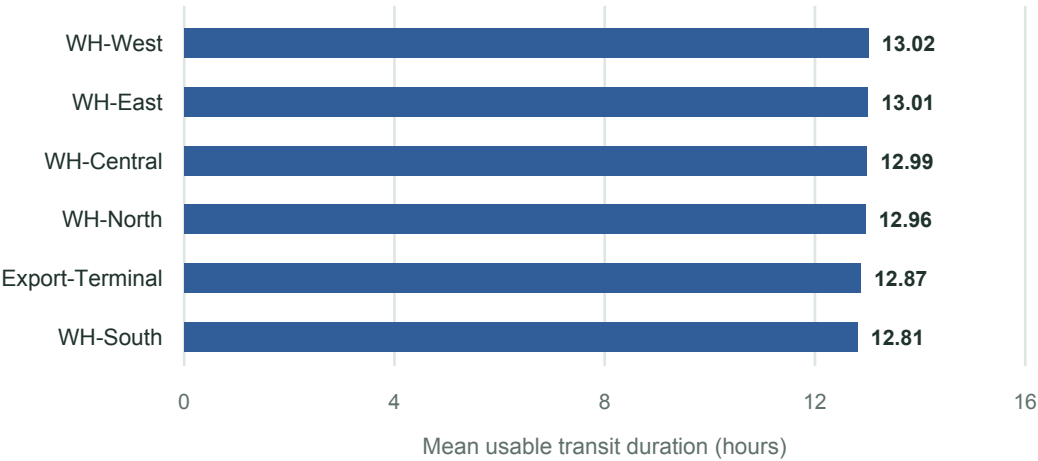


Figure 6. Warehouse means from Query 7, including labelled reported fallbacks. Full trip log; destination-specific dates are not supplied in this aggregate.

- Finding:** Warehouse means range only from 12.81 hours at WH-South to 13.02 at WH-West; their medians range from 12.90 to 13.20 hours.
- Meaning:** Review individual routes and distance context before interpreting small destination-level differences as performance gaps.
- Limitation:** No SLA or promised arrival exists, and 2,356 of 9,715 usable durations are reported fallbacks rather than timestamp calculations.

Selected route	Trips	Median h	Km	Fallback
MANDI051 / Export-Terminal	21	18.30	798	19.05%
MANDI033 / WH-West	24	18.05	785	54.17%
MANDI051 / WH-North	24	17.60	751	20.83%

Table 2. Top three of ten routes from Query 8; minimum 15 usable trips. Km is mean distance for the usable-duration population.

- Route finding:** Gulbarga Market (MANDI051) to Export-Terminal leads at an 18.30-hour median over 21 trips and 798.30 km mean distance; its calculated-only median is also 18.30 hours.
- Review implication:** South Dumdum Grain Market (MANDI033) to WH-West has an 18.05-hour median, but 54.17% of its 24 trips use reported fallback durations; its calculated-only median is 17.70 hours.

Long duration is not confirmed delay. No route-delay-frequency ranking or warehouse crop-volume estimate is inferred.

Conclusions and recommended actions

Conclusions

Supply is distributed across six reporting crops and many mandis: the largest crop contributes 17.19%, and the five largest mandis contribute 9.62%. August leads the full-month totals, but commodity-specific patterns differ.

Price exposure is broad: 40.16% of eligible observations are below supplied MSP, despite positive average gaps for every crop. Monitoring the mean alone would hide this distributional risk. Mandi-crop screening identifies specific review candidates.

Warehouse means are tightly clustered, while selected route medians reach 18.30 hours. Duration source, trip distance and small group sizes must accompany any operational interpretation.

Evidence	Recommended action
Balanced crop shares; diffuse mandi volumes	Plan network-wide handling capacity; review monthly crop mix before reallocating resources.
Widespread below-benchmark observations	Track rates, eligible counts and coverage together; investigate shortlisted mandi-crop combinations.
Long routes and reported fallbacks	Review route records and improve departure/arrival capture; establish an SLA before measuring delays.
Conflicting locations and date assumptions	Maintain a governed reference table and record original timestamps with explicit timezone labels.

Limits and reproducibility

Results are descriptive, not causal or representative of real markets. September is partial; crop and price averages do not measure revenue, losses or unique farmers. Source/master location precedence is a rule, not independent geographical verification. Weather mapping and rainfall intervals remain unresolved, so no weather-causation claims are included.

The supplied SQL arrival aggregates differ from the earlier Python control by 131.86 qtl; the difference is disclosed rather than silently reconciled. No live database was rerun for this report: analysis uses the eight supplied CSV result sets and inspected MySQL queries. Charts reproduce those aggregates, not unverified dashboard filters.

Evidence index: Q1 crop distribution; Q2 monthly arrivals; Q3 top mandis; Q4 crop price risk; Q5 mandi-crop risk; Q6 monthly prices; Q7 warehouse transit; Q8 long routes. Selected tables are abridged; SQL and full outputs are maintained separately.

Explore the complete interactive dashboard: <https://mandhidashbord.netlify.app/>